

Multiple Choice

1. **Answer:** A
Options: A) 8 over 12; B) 9 over 6; C) 2 over 6; D) 24 over 27
Note: Correct option: A - 8 over 12
2. **Answer:** A
Options: A) 298; B) 126; C) 592; D) 396
Note: Correct option: A - 298
3. **Answer:** C
Options: A) 12; B) 1; C) 30; D) 5
Note: Correct option: C - 30
4. **Answer:** D
Options: A) 5 over 3; B) 3 over 8; C) 2 over 5; D) 3 over 5
Note: Correct option: D - 3 over 5
5. **Answer:** D
Options: A) 16; B) $2\sqrt{2}$; C) 5; D) $5+2\sqrt{13}$
Note: Correct option: D - $5+2\sqrt{13}$

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '5'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '3.8, 3.18, 3 and 1 over 8, 3'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** In the transition between figures, each endpoint splits into two new segments, creating two new endpoints, so the number of endpoints doubles. Figure 1 has 3 endpoints, so Figure n has $3 * 2^{n-1}$ endpoints. Thus Figure 5 has 48 endpoints.
Note: Students should show all steps in their mathematical solution.
12. **Answer:** -128. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.